

CPEN 438 – PROJECT 13 | GROUP 1 | WEEK 3

RESPONSIBLE AI-USE DECLARATION

Project: Wiring the Data Centre: Interconnection Networks for a Simulated Accra Warehouse-Scale Cluster

Team	Group 1	Reporting Week	Week 3
AI tool used	Claude (Anthropic)	Dates of use	23 August 2026

1. Declaration principle

AI was used as a supporting tool at specific, bounded points in the week's work. No result or implementation claim was accepted because an AI tool produced it. Every figure, table and numeric statement in the Week 3 submission was produced by code that the team wrote or reviewed line by line, compiled locally with warnings enabled, and ran; each was then cross-checked against the raw result files committed to the repository. The team understands, can reproduce, and can defend all submitted material.

2. Scope of AI use

AI assistance was confined to four areas: checking our reading of the brief against the Week 1 and Week 2 artefacts, reviewing and debugging specific C functions that we had already drafted, suggesting library calls and plotting idioms for the Python analysis stage, and improving the wording of report prose that we had already outlined. The parts of the week that carry the marks were done by the team.

Specifically, AI was NOT used to choose the network model or its parameters, to decide the experimental method, to interpret any measured result, or to reach any conclusion presented in the design specification. The three substantive corrections made during the week — diagnosing the ring deadlock, rejecting the biased random-number generator, and redesigning the innovation after its first version failed — were all made by team members reading output and reasoning about it, and each is recorded in the weekly report.

3. Work completed by the team without AI assistance

- The choice of network model: packet and flit sizes, buffer depths, router delay, credit-based wormhole flow control, and the two-phase per-cycle evaluation that prevents a flit crossing two routers in one cycle.
- Diagnosis of the ring deadlock. The hang at high offered load was traced to the cyclic channel-dependency graph of a bidirectional ring under shortest-direction routing, and resolved with the dateline virtual-channel rule from Dally and Towles.
- The decision to give both topologies the same virtual-channel budget, so that the ring-versus-mesh comparison measures topology rather than buffer allocation.
- The ejection-bandwidth assumption and its justification, which materially determines the hot-node results and is stated explicitly in §4.4 of the design specification.

- Rejection of the first random-number generator after review showed it biased the Bernoulli injection test, and the decision to require bit-for-bit equality between the C and Python generators.
- The design of the innovation, including the diagnosis that the first virtual-channel-only version failed because the shared source queue still blocked, and the decision to add two controls and hold total queue capacity constant.
- The experimental method: the sweep range, the warmup, measurement and drain protocol, and the throughput-divergence saturation criterion.
- All interpretation of results, including the finding that hot-node traffic moves the binding constraint off the bisection cut, and the explanation of why the analytical model over-predicts saturation under uniform traffic.

4. AI use record

Each row records one bounded use of AI assistance during the week.

Date	Prompt/query summary	Output received (summary)	How verified	What was modified by the student/team	Final sections/files containing AI-assisted material
23 Aug 2026	Check our reading of the Week 3 scope against the Project 13 brief, the four-week schedule and our Week 1 and Week 2 artefacts.	A checklist of Week 3 requirements and their dependencies on earlier weeks.	Compared line by line with the CPEN 438 master brief and the Week 2 design specification.	Items the brief does not require were removed. The team decided independently to defer the folded torus and adaptive routing to Week 4 rather than claim them early.	Design specification §1 and §2; weekly report §1.
23 Aug 2026	Review C functions we had drafted for the flit-level simulator, and help debug the virtual-channel allocation and switch-arbitration logic.	Suggested corrections to the arbitration loop bounds and to the release of a virtual channel when a tail flit departs.	Compiled with GCC using -O2 -Wall -Wextra -std=c11 with no warnings. The Week 3 suite returned ALL TESTS PASSED (97 checks across 12 test functions). The packet-conservation identity was asserted at all 216 sweep points with zero violations.	The team wrote the model, the pipeline structure and the deadlock fix. Suggested changes were read, tested and in two cases rejected before the remainder were accepted.	student_implementation/network_sim.c and .h.
23 Aug 2026	Help implement the xorshift64* and SplitMix64 arithmetic identically in C and in Python, once the team had chosen that generator.	The shift-and-multiply sequence and the 64-bit masking needed for the Python side to match C.	tests/dump_traffic.c and the Python --verify mode confirmed 3000/3000 matching destinations for both patterns. Uniformity was measured flat to within 8% over 200 000 draws.	The choice of generator, the seed-derived hot-node rule and the fall-through rule for a hot node drawing itself were the team's design decisions and were implemented before any assistance was sought.	student_implementation/traffic.c and .h; python/gen_datacenter_traffic.py.
23 Aug 2026	Suggest Matplotlib idioms for the saturation-curve figures and help translate our channel-load model into MATLAB syntax.	Plotting boilerplate for the seven figures and MATLAB equivalents for the path-walk loops.	The model was checked against the Week 2 closed forms: it reproduces diameter 8/6 and bisection 2/4 independently. Predictions were compared point by point with the measured sweep.	The model itself — the channel-load walk, the ejection cap and the M/D/1 queueing term — was derived by the team from Dally and Towles. Axis choices, the efficiency column and the decision to report the classical bisection bound alongside were the team's.	matlab/bisection_latency_model.m; python/analyze_network_results.py.
23 Aug 2026	Improve the wording and structure of report prose the team had already outlined and populated with results.	Edited paragraphs and table layouts for the weekly report and design specification.	Every numeric claim was re-checked against results/processed/week3_results_table.csv, week3_model_vs_measured.csv and week3_innovation_table.csv before submission.	All technical content, conclusions and limitations were written by the team. No commit hash was entered that had not been read from the repository log; unverified hashes are shown as placeholders.	Week 3 weekly report; Week 3 design specification.

5. Human verification and evidence

Verification item	Evidence checked	Result
Assigned configuration	configs/group1_week3_config.txt	16 nodes; 16-node ring; 4×4 mesh; shortest-direction ring routing; dimension-order mesh routing; seed 1301; hot database nodes 5 and 1 derived from the seed; 50% hot share.
Week 2 regression	Rebuilt test_topology from repository source	ALL TESTS PASSED (35 checks across 7 test functions). Ring 8/2 and mesh 6/4, unchanged.
Week 3 unit tests	Rebuilt test_simulator from repository source	ALL TESTS PASSED (97 checks across 12 test functions).
Packet accounting	run_sweep console output	0 conservation violations across 216 runs; 0 packets dropped at or below the reported saturation rate for every configuration.
Traffic-generator equivalence	dump_traffic compared with gen_datacenter_traffic.py --verify	uniform 3000/3000 and hotnode 3000/3000 destinations match; the two generators are bit-for-bit identical.
Saturation results	results/processed/week3_results_table.csv	Uniform: ring 0.28, mesh 0.70 flits/node/cycle. Hot-node: 0.20 for both.
Model validation	results/processed/week3_model_vs_measured.csv	Efficiency against the ideal channel-load bound: 0.67 and 0.75 under uniform traffic; 0.84 and 0.91 under hot-node traffic.
Innovation ablation	results/processed/week3_innovation_table.csv	At an offered load of 0.24 on the mesh, background latency falls from 574.0 to 8.0 cycles. Neither control reaches this: 387.3 with extra buffers alone, 157.4 with class queues alone.
Scope control	Project 13 four-week schedule	No folded-torus, adaptive-routing or Week 4 results are claimed.
Independent rebuild	Clean clone of the repository rebuilt by a second team member	All five executables build with no warnings; both test suites pass; hand-derived channel-load bounds for the hot-node case match the values the MATLAB model reports.
Git history	Repository commit log	Week 3 commits are listed by message in design specification §14.2; hashes are inserted from the log after push.

6. Modifications and human responsibility

- All AI-assisted code was reviewed, edited and tested before use. The Week 3 code was compiled locally with warnings enabled and verified with automated tests, an end-to-end packet-conservation check, and comparison against an independently derived analytical model.
- Suggested changes were not accepted uncritically. Two were rejected during review of the simulator, and the first version of the innovation was discarded entirely after the team established why it did not work.
- Modelling assumptions that materially affect the results, in particular the ejection bandwidth used under hot-node traffic, are stated explicitly in the design specification rather than left implicit in the code.

- Where the analytical model disagrees with measurement, the disagreement is reported and explained rather than adjusted away.
- The innovation is reported with its limits: the hot flow itself is not made faster, and on the ring the loss-free saturation rate is unchanged. No claim is made beyond what the measurements support.

7. Responsibility and confirmation

Group 1 accepts responsibility for the correctness of all submitted material, including sections where AI assistance was used. Every team member can explain the design decisions, the experimental method and the results recorded above.

Team Leader: Bernardine Adusei-Okrah

Date: 23 August 2026

Team Members' Initials: BAO, PES, VEA, MNHABM, FS